

MSDS 520 Lecture Notes

Vectors and Matrices

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1 Introduction

1.1 Motivation

Why study linear algebra?

- It provides a framework for solving **systems of linear equations**.
- It describes the **geometry of space**: directions, distances, projections.
- It is the mathematical foundation for **machine learning and deep learning**.
- Real-world data science applications include regression, principal component analysis (PCA), and network analysis.

Emphasis: We will focus both on *geometric intuition* (seeing what vectors and matrices *do*) and *applications* in data science.

1.2 From Systems to Vectors and Matrices

In the previous unit, we studied how to solve systems of linear equations using substitution, elimination, and Gaussian elimination. A key idea was rewriting a system in compact matrix form. For example:

$$\begin{cases} 2x + y = 5 \\ 3x - y = 4 \end{cases} \longleftrightarrow \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}.$$

Here:

- The **matrix** stores the coefficients of the system.
- The **vector of unknowns** stores the variables.

- The **vector of constants** stores the right-hand sides.

Gaussian elimination allowed us to manipulate these arrays systematically until the solution appeared.

Now, we step back: rather than just seeing vectors as “columns of variables” and matrices as “tables of coefficients,” we will study them as mathematical objects in their own right:

- Vectors describe points, directions, and displacements in space.
- Matrices describe transformations such as rotations, scalings, and reflections.
- This broader perspective connects algebra to geometry and powers the linear algebra tools used in machine learning and data science.

2 Vectors

2.1 Definition and Notation

A **vector** is a quantity with both magnitude and direction.

Notation:

- Written in **boldface**: $\mathbf{v}$.
- **Column vector** with components:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

- **Row vector** (the transpose of a column vector):

$$\mathbf{v}^T = [v_1 \quad v_2 \quad \cdots \quad v_n].$$

Unit vectors: vectors of length 1, giving pure direction (e.g. $\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in 2D).

2.2 Vector Operations

- Addition:

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix}.$$

- Scalar multiplication:

$$c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}.$$

- Linear combinations: $a\mathbf{v} + b\mathbf{w}$.

Span: The set of all linear combinations of a collection of vectors. Interpretation: directions we can reach by combining vectors.

2.3 Inner Product, Norms, Distances, and Angles

For $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$\mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^n v_i w_i.$$

Matrix notation: inner product is row $\times$ column:

$$\mathbf{v} \cdot \mathbf{w} = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}.$$

- **Norm:** $\|\mathbf{v}\| = \sqrt{\sum v_i^2}$.

- **Distance:** $\|\mathbf{v} - \mathbf{w}\|$.

- **Angle:**

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}.$$

- **Projection:**

$$\text{proj}_{\mathbf{w}}(\mathbf{v}) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{w}\|^2} \mathbf{w}.$$

3 Linear Independence and Basis

3.1 Linear Dependence

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly dependent** if one can be expressed as a combination of the others. Intuition: redundancy of directions.

Connection: in regression, collinearity among predictors.

3.2 Basis and Dimension

- A **basis** is a minimal set of independent vectors spanning the space.
- **Dimension:** number of vectors in the basis.
- Orthonormal bases: mutually orthogonal, unit length vectors.

4 Matrices

4.1 Definition and Notation

A **matrix** is a rectangular array of numbers.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

Special matrices:

- Zero: $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.
- Identity: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- Diagonal: $\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$.
- Symmetric: $\begin{bmatrix} 1 & 4 \\ 4 & 3 \end{bmatrix}$.

4.2 Basic Arithmetic

- Addition: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$.
- Scalar multiplication: $2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$.
- Transpose: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$.

4.3 Matrix-Vector Multiplication

Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$:

$$A\mathbf{v} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Interpretation: applying A to $\mathbf{v}$ *transforms* it, changing its length and direction. Geometrically: stretching, rotating, reflecting.

5 Matrix Examples

5.1 Geometric Transformations

- Rotation by θ : $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.
- Reflection across x -axis: $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.
- Scaling by 2 in x and 3 in y : $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$.

5.2 Incidence Matrices

A graph with edges $\{(1, 2), (2, 3)\}$ has incidence matrix:

$$B = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}.$$

Each row encodes an edge, with $+1$ at the start node and -1 at the end.

5.3 Social Networks Example

An adjacency matrix for 3 people, where person 1 is friends with 2 and 3, and 2 is friends with 3:

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Row sums (degrees) give the number of connections. A^2 counts the number of two-step connections.

6 Matrix Multiplication

6.1 Matrix-Matrix Multiplication

Example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Properties:

- Not commutative: $AB \neq BA$ in general.
- Associative: $(AB)C = A(BC)$.

6.2 Matrix Powers

$A^2 = A \cdot A$, useful in repeated transformations or Markov chains.

7 Matrix Inverses

7.1 Left, Right, and Complete Inverses

For A (possibly rectangular):

- A **left inverse** B satisfies $BA = I$.
- A **right inverse** C satisfies $AC = I$.
- A **two-sided inverse** A^{-1} satisfies both $AA^{-1} = I$ and $A^{-1}A = I$ (only possible when A is square and nonsingular).

7.2 Determinants

For 2×2 :

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

For 3×3 :

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Interpretation: scaling factor for area/volume.

7.3 Computing Inverses

For 2×2 :

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

For larger matrices: Gaussian elimination or numerical methods.

7.4 Pseudo-inverse

The **Moore-Penrose pseudo-inverse** A^+ generalizes the inverse to rectangular or singular matrices. Applications: regression normal equations $\hat{\beta} = X^+y$.

8 Eigenvalues and Eigenvectors

8.1 Definition

$\mathbf{v}$ is an eigenvector of A if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

8.2 Computation

Characteristic polynomial: $\det(A - \lambda I) = 0$.

8.3 Properties

- Eigenvalues may be real or complex.
- Symmetric A : eigenvectors are orthogonal.
- Diagonalization: $A = VDV^{-1}$.

8.4 Applications

- Stability of systems: powers of A .
- PCA: eigenvectors of covariance matrix give principal directions.

Concluding Notes

This set of notes provides the conceptual and computational foundation for linear algebra with a focus on data science applications. Students should focus on both the **geometry** (what transformations do) and the **algebra** (how to compute).